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Agent Reshapes the New Paradigm of the Software Industry

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Jason Li
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Chairman of Singularity Intelligence Technology Summit (SITS), and member of the ISO-C++ Standard Committee. He has rich experience and in-depth research in artificial intelligence, software architecture and product innovation. In recent years, he has mainly focused on the application of artificial intelligence methods centered around Large Language Models. He proposed the "ParaShift Cube" for technological product innovation. He is also a serial entrepreneur who has founded well-known technology companies including SoftCompass, SlideIdea, and Boolan.

Agentic Scaling : the 4th Paradigm of Scaling Law

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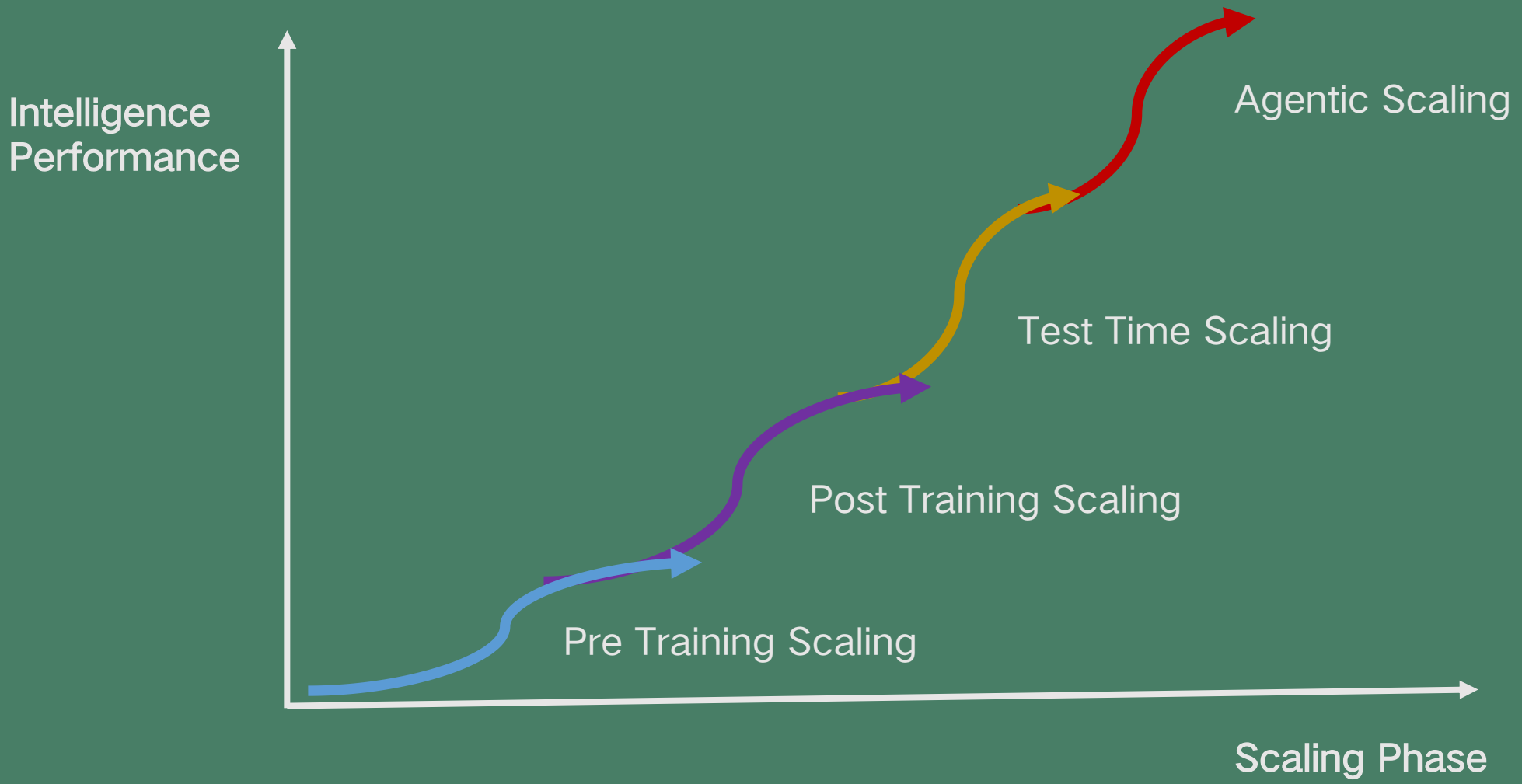

Agentic Scaling : the 4th Paradigm of Scaling Law

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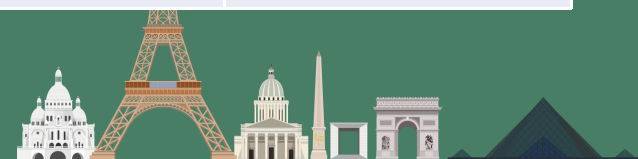
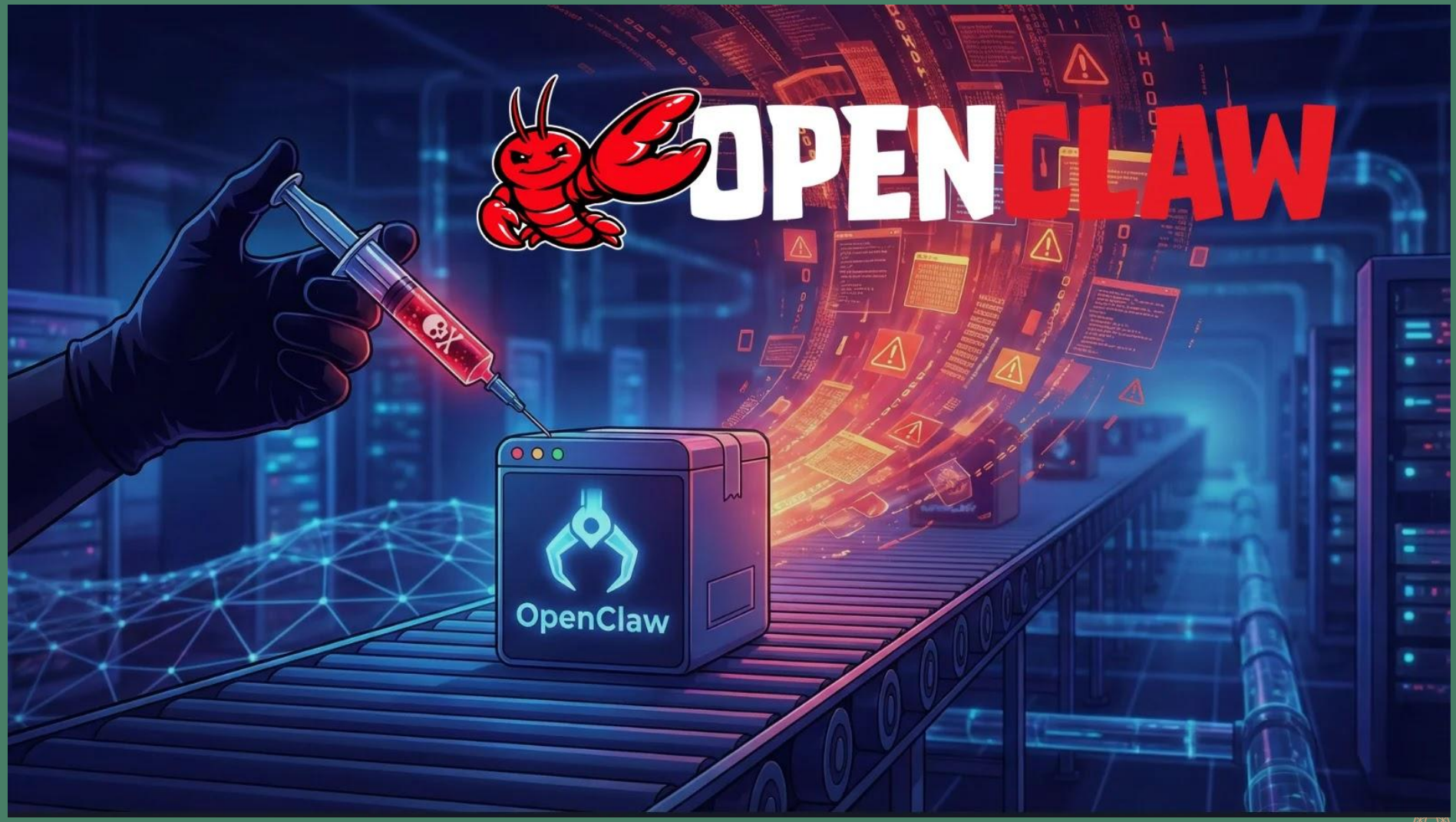
Scaling Paradigm	Core Mechanism	Computation Phase	Scaling Dimension	Scaling Objective
Pre-training Scaling	Larger models + More data	Training phase	Parameter count / Data volume	Representation compression
Post-training Scaling	RLHF、SFT	Post-training fine-tuning phase	Preference data / Reward signal strength	Behavior Alignment
Test-time Scaling	Inference-time thinking (COT)	Single inference process	Number of thinking steps / Sampling width	Depth of reasoning
Agentic Scaling	**Multi-step, cross-system, self-loop execution**	**Continuously action sequences at runtime**	**Number of agents × Iteration depth × Time span**	**Task completion rate**



Crazy OpenClaw



Every generation of technology revolution brings dual disruption to developers.

1. Disrupts the form of applications: **What to DO**

2. Disrupts the way of development : **How to**

What changes do Agents bring to software?

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	Internet	AI
Application Paradigm	Web (HTML/CSS/JS)	AI Agent
Development Paradigm	Cloud Native	AI Native

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Agent Changes Software Application Paradigm

Agent Changes Application Paradigm

User Interface	Agents will replace traditional software as the primary interface for human interaction.
Downstream Tools	Traditional software will move downstream, becoming tools invoked by Agents.
Design Refactoring	Design software for Agents, not for Humans. CLI is more Agent-friendly than GUI.

The Evolution of Interaction Driven by Agents

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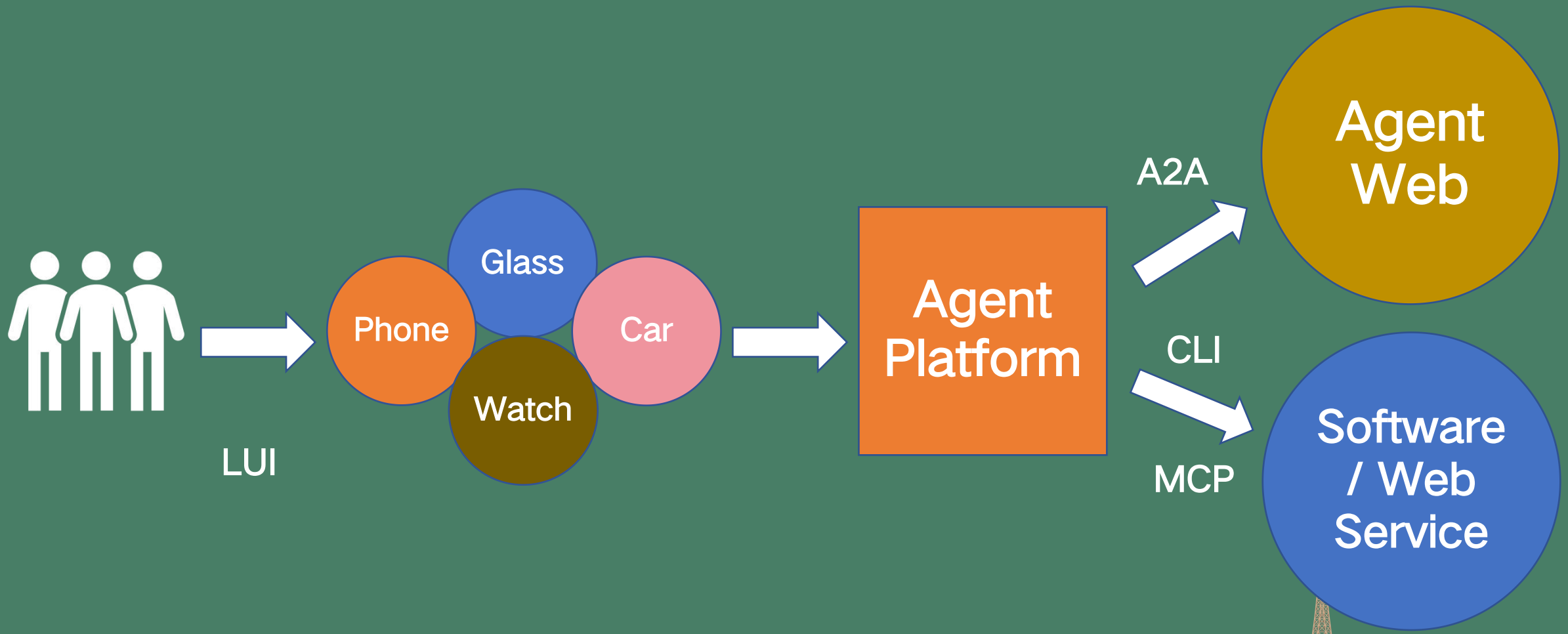

Design software for Agents, not for Humans

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Tools

Build the software tools and APIs required for the Agent's execution chain.

Skills

Develop task-specific skills for a dedicated Agent.

Skills = Job Description + SOP + Toolkit

Environment

Create environment (cloud/client) for Agent, with sandbox permissions.

Environment = System + Permissions

The Evolution of the Software Spectrum

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Stable
Design for Reuse

Stable Core +
Elastic Extensions

Instantly Generated,
Disposable after Use

A Comparison of Three Software Paradigms

	Traditional Software	Elastic Software	Instant Software
Reuse Form	Stable source code (binary)	Stable source code (binary) + Skills	Skills
Software Engineering	Follows traditional SDLC engineering	Traditional SDLC system + Runtime engineering system	Abandons SDLC, adopts runtime engineering system
AI Development	Spec-driven	Spec + Vibe Coding	Vibe Coding
Capability Binding	Development phase: full definition & implementation Runtime phase: executes pre-compiled deterministic rules	Development phase: stable, high-certainty core capabilities Runtime phase: highly variable, context-dependent capabilities dynamically generated by Agents	Runtime phase: dynamically generated and executed by Agents based on tasks and Skills

Key Characteristics of Disruptive Innovation

- Instant Software refers to software that is automatically generated by Agents, or "black-box created" by ordinary users using "natural language programming".

- Just like Web programming in the Internet era, Instant Software represents an incremental market and creates a new paradigm for software in the AI era.

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Agent Changes Software Development Paradigm

Agent Changes Development Paradigm

Programming → AI Copilot → Agentic Coding

AI-Native Software Development Maturity Model

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DIMENSION	L1 Assisted Productivity Unlock Individual Potential	L2 Domain Integration Build Organizational Intelligence	L3 Agent Collaboration Multiply Team Effectiveness	L4 Autonomous Agents Enable Autonomous Delivery	L5 Self-Evolving Factory Drive Continuous Evolution
Infrastructure	• Call foundation models through **public cloud APIs** • Traditional cloud resources, no dedicated `AI Infra` • Lack of a unified model management platform	• **Hybrid deployment** with privatized domain models • Dedicated `GPU / NPU` compute resource pool • Digitized storage of domain knowledge • Unified `API Gateway`	• **Agent runtime environment** and version management • Long-running Agent session management • Domain `Harness` configuration management platform	• **Multi-Agent autonomous orchestration platform** and scheduling engine • `AgentOps` decision-chain tracing and token monitoring • **Token optimization** infrastructure (Caching / Routing)	• **Self-evolving flywheel infrastructure**: automated evaluation and online optimization feedback loop • Intelligent **cost-benefit optimization platform** • Adaptive resource elasticity and cost optimization engine
Knowledge Engineering	• Use only public knowledge, with **no domain integration** • Documents are unstructured; retrieval is mainly manual • No engineered knowledge management approach	• **Domain knowledge base** development • `Context Engineering` practices initiated • Case library and experience-capture mechanism	• Domain `Harness` knowledge system construction • Domain **Skill library** and quality management system • Real-time expert knowledge injection and lesson capture	• **Knowledge flywheel**: multi-Agent sharing and contribution • Agents possess **manager-level** knowledge synthesis capability • Agent self-optimization experience is automatically captured	• **Self-evolving knowledge ecosystem**: autonomously identifies blind spots and expands itself • Cross-task knowledge abstraction and hierarchical domain knowledge systems • **Automatic conversion** from knowledge to capability and innovation generation
Process & Tools	• Embed **AI-assisted tools** into traditional workflows • Processes are human-led; AI tools are optional • `Vibe Coding` used exploratorily	• AI integrated into key `DevOps` stages • **Intelligent code review** and test generation • AI-based code quality baselines and technical debt management	• Adaptive process standards (`Spec-driven`) • Domain **Harness Engineering** practices • Agents take on explicit R&D roles; performance and token efficiency are measured	• `AgentOps` and `VPT` metrics drive continuous optimization • Agents possess **autonomous design and planning capabilities** • Multi-Agent autonomous orchestration, collaboration, and token optimization	• **End-to-end delivery flywheel pipeline** • Four-layer flywheel path: single task → cross-task → end-to-end → self-evolution • Flywheel self-evolution and automatic discovery of configuration blind spots
Organization & Talent	• Traditional organization plus **AI tool training** • Basic `Prompt Engineering` skills • Individual exploration, with no systematic capability building	• `AI CoE` rollout practices and training system • Developers learn **Context Engineering** • AI usage efficiency is included in measurement	• Role transformation: **Harness Engineer**, etc. • Developers transition into **Agent collaboration supervisors** • Human-AI collaboration norms and evaluation standards	• **Agent Orchestrator / Evaluator** as new roles • Organizational operations approach an **automated software factory** • Evaluation based on value creation and token efficiency	• **Flywheel Architect**: designer and guardian of the flywheel • Networked ecosystem organization where humans and AI evolve together • Innovation incubation platform: complete production and innovation teams formed by **micro-teams (super-individuals) + multi-Agent systems**
Security & Governance	• Basic AI usage rules and prohibited items • AI content subject to **manual review** • Simple logging, with no systematic risk assessment	• **Governance framework** including access control and audit • `AI Gateway` and data masking • AI code security scanning and compliance checks	• **Agent permission boundaries** defined and enforced • Harness-layer **security design** and sandbox isolation • Complete traceable audit of Agent behavior	• **Permission matrix** and monitoring under autonomous orchestration scenarios • **Cognitive debt** assessment and control • Security constraints for Agent self-optimization and **automatic rollback**	• **Autonomous governance ecosystem**, with AI participating in supervision • AI trustworthiness rating and impact assessment • **Human-AI co-governance**, with humans retaining ultimate control

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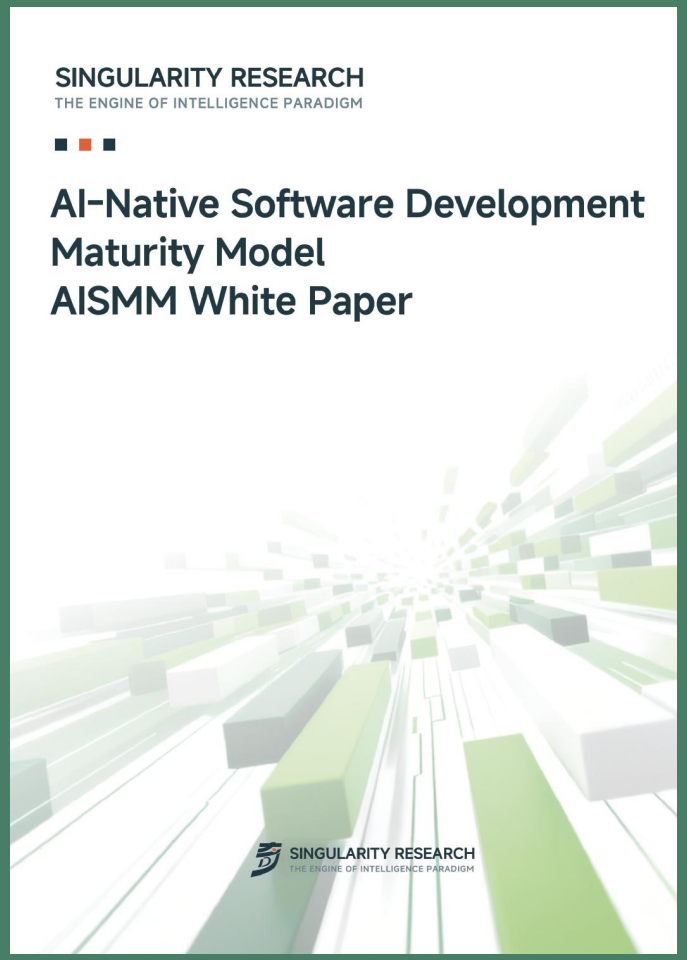

AI-Native Software Development Maturity Model

AISMM White Paper

Released by Singularity Research

Five Stages of AISMM Maturity

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Level 1	Level 2	Level 3	Level 4	Level 5
AI Assisted	Domain Integrated	Agent Synergy	Autonomous Agents	Self-Evolving Factory
Unlock individual potential	Build Organizational Intelligence	Multiply Team Effectiveness	Enable Autonomous Delivery	Drive Continuous Evolution

Five Elements of the AISMM Maturity Model

1. AI Software Infra
2. Knowledge Engineering
3. Process & Tools
4. Organization & Talent
5. Governance & Security

Human in the Loop

Human out of the Loop

Human-Agent Synergy

Autonomous Agents

Harness Engineering

Prompt Engineering

（2022-2023）

Focus on how to clearly articulate requirements to the model in multi-turn dialogues.

Context Engineering

（2024-2025）

Focus on how to provide just the right information in long-horizon human-AI collaborative tasks.

Harness Engineering

（2026-...）

Focus on how to build a closed-loop system that enables Agents to perform tasks reliably, safely and collaboratively.

The Core Elements of Harness Engineering

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Know

- ✓ Context
- ✓ Memory
- ✓ Skills

Action

- ✓ Tools
- ✓ Infra
- ✓ Orchestration

Feedback

- ✓ Evaluation & Validation
- ✓ Debugging / Correction
- ✓ Tracking / Observability

Manual Orchestration	AI-Assisted	Autonomous Optimization
OpenClaw	*Hermes Agent*	*AutoHarness*

Deep Restructuring of Cloud Computing Service Models

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IaaS -> Token Infra

IaaS is evolving from hardware resource leasing into infrastructure services for Token factory

PaaS -> MaaS

PaaS is transforming from an application runtime platform into a composable AI capability platform: MaaS + Agent Runtime.

SaaS -> AaaS

SaaS is evolving from a function delivery platform into an agent service platform centered on task delivery — Agent-as-a-Service (AaaS).

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12 Frontier Trends of AI Industry Transformation in 2026

Application Paradigm: Natural Language Interaction Reshapes New Software Forms

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Trend 01. Agents Become the First User Entry Point, While Traditional Software Moves Downstream

Trend 02. Generative UI (GenUI) Creates More Delightful Personalized User Experiences

Trend 03. Agents Refactor the Internet from an Information Network into an Action Network

Trend 04. Natural Language + Agent-Centric Networks Become the Human-Machine Interface Layer Across Devices

Development Paradigm: From Code Generation to Flywheel Construction

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Trend 05. Instant Software Grows Rapidly, While Elastic Software Reaches Engineering Maturity

Trend 06. Harnesses Mature from Manual Configuration to AI-Assisted Generation and Automated Optimization

Trend 07. Development Tools and Infrastructure Become Agent-Native, Not Human-Centric

Trend 08. The Boundary of Autonomous Software Engineering Keeps Expanding

Computing Paradigm: Token is Restructuring AI Infrastructure

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Trend 09. Deep Restructuring of Cloud Service Models—from IaaS/PaaS/SaaS to Token Factory/MaaS/AaaS

Trend 10. Token Economics Becomes Infrastructure Engineering, Not Application-Layer Hacking

Trend 11. Pooling, Heterogeneity, and Elastic Scheduling of Inference Compute

Trend 12. Edge-Cloud Collaborative AI Architectures Reach Engineering Maturity

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Thanks!

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